Hybrid Recommendation Framework for Intelligent Service Provider Matching

***Abstract*—Finding suitable service providers for customer requirements remains a significant challenge in existing online service marketplaces. Most platforms rely on basic keyword search and manual browsing, often returning large numbers of providers without considering relevance, semantic meaning, service quality, or user preferences. As a result, users spend considerable time evaluating providers, while highly qualified service providers may not receive appropriate visibility. To address this issue, this research proposes a Hybrid Service Provider Recommendation Framework that combines Content-Based Filtering, Deep Learning-based Semantic Matching, and Collaborative Filtering techniques. The framework integrates TF-IDF for keyword relevance analysis, Multilingual BERT for semantic similarity understanding, and a novel Credibility-Based Collaborative Filtering algorithm that incorporates provider ratings, booking success rates, interaction frequency, and user interaction history. The proposed system utilizes three datasets: a Provider Dataset containing service provider profiles, a User Request Dataset containing customer requirements, and a User Interaction Dataset containing booking and rating histories. Recommendation scores generated by the three modules are normalized and combined using a weighted hybrid scoring mechanism. Experimental evaluation was conducted using Precision@K, Recall@K, and NDCG@K metrics to assess recommendation quality. Results demonstrate that the proposed hybrid approach consistently outperforms individual recommendation techniques by improving ranking accuracy and relevance. The framework provides a scalable and intelligent solution for service provider recommendation systems and can be extended to various service-oriented digital marketplaces.**

***Keywords*—** Hybrid Recommendation System, Service Provider Recommendation, TF-IDF, Multilingual BERT, Collaborative Filtering, Semantic Similarity, Information Retrieval, Machine Learning.

I. Introduction

Online service marketplaces have become increasingly popular by connecting customers with professionals across various service domains, including plumbing, electrical work, carpentry, masonry, air-conditioning, cleaning, painting, CCTV installation, and vehicle repair. The rapid growth of these platforms has significantly improved service accessibility by enabling users to search and hire providers through digital marketplaces. However, identifying the most suitable provider for a specific customer requirement remains a challenging task [4], [5].

Most existing service marketplaces rely on keyword-based search, category filtering, and provider ratings for recommendations. However, these approaches often fail to capture the semantic meaning of user requests or provide personalized recommendations [3]. As a result, users must manually compare multiple providers based on ratings, experience, and service descriptions.

In addition, keyword matching alone cannot recognize semantically similar requests expressed using different words. For example, user queries such as ”repair leaking water pipe” and ”fix plumbing issue” describe the same service requirement but may produce different search results when only lexical similarity is considered [1], [2], [6]. Similarly, recommendation systems that depend solely on collaborative filtering suffer from the cold-start problem and limited interaction histories, making them less effective for new users or providers [7]–[9]. These limitations reduce recommendation accuracy and negatively affect user satisfaction.

To overcome these challenges, this research proposes a Hybrid Recommendation Framework for Intelligent Service Provider Matching. The proposed framework integrates three complementary recommendation techniques: TF-IDF-based content matching, Multilingual BERT-based semantic similarity analysis, and a custom Credibility-Based Collaborative Filtering algorithm [1], [2], [4]. The TF-IDF module captures explicit keyword relevance between user requests and provider descriptions, while the Multilingual BERT module identifies contextual semantic relationships beyond exact keyword matching [1], [2]. The proposed collaborative filtering module further improves personalization by incorporating provider credibility, booking success rate, interaction frequency, and historical user interactions [8], [9].

Unlike conventional recommendation systems, the proposed framework combines the outputs of these three modules using a weighted hybrid score fusion strategy to generate more reliable provider rankings [4], [5]. The framework utilizes three datasets, namely a Provider Dataset, a User Request Dataset, and a User Interaction Dataset, to support recommendation generation and personalization.

The primary contribution of this work is the development of a credibility-aware collaborative filtering algorithm that extends traditional collaborative filtering by integrating provider quality indicators into the recommendation process. Experimental evaluation conducted using Precision@K, Recall@K, and Normalized Discounted Cumulative Gain (NDCG@K)

demonstrates that the proposed hybrid framework consistently outperforms individual recommendation approaches [11]. The proposed system therefore provides an accurate, scalable, and intelligent recommendation solution suitable for large-scale online service provider marketplaces.

II. Literature Review

Recommendation systems have become a fundamental component of modern online service platforms by assisting users in identifying the most appropriate services from a large pool of available providers. Traditional content-based recommendation techniques primarily rely on textual similarity between user requests and service descriptions. Among these approaches, the Term Frequency–Inverse Document Frequency (TF-IDF) weighting scheme has been extensively adopted due to its simplicity, computational efficiency, and effectiveness in measuring keyword relevance within textual documents [3]. However, TF-IDF relies solely on lexical similarity and cannot capture semantic relationships between different expressions describing the same service.

Recent advances in Natural Language Processing (NLP) have significantly enhanced semantic understanding through transformer-based language models. Bidirectional Encoder Representations from Transformers (BERT) introduced contextual word representations by considering both preceding and succeeding words during language modeling, enabling superior semantic understanding compared to traditional embedding techniques [1]. Subsequently, Sentence-BERT (SBERT) extended the original BERT architecture by generating fixed-length sentence embeddings suitable for efficient semantic similarity computation while significantly reducing computational complexity [2]. These models have become widely adopted in semantic search and recommendation systems because they effectively identify contextual similarities even when different vocabularies are used to describe identical service requirements [6], [13].

Collaborative Filtering (CF) is one of the most widely used recommendation techniques for generating personalized recommendations based on historical user interactions. Matrix Factorization and Factorization Machine models have demonstrated strong performance in discovering latent relationships between users and items using interaction data [7], [8]. Despite their effectiveness, conventional collaborative filtering approaches heavily depend on sufficient interaction history and frequently suffer from the cold-start problem when new users or providers have limited historical information. Furthermore, most existing collaborative filtering models do not explicitly consider provider credibility or service quality indicators during recommendation generation [5], [14].

To overcome the limitations of individual recommendation techniques, several hybrid recommendation systems have been proposed by combining content-based filtering with collaborative filtering. Hybrid approaches improve recommendation accuracy, robustness, and personalization by exploiting complementary strengths of multiple recommendation strategies [4], [5]. Recent studies have further demonstrated that hybrid recommendation frameworks consistently outperform standalone recommendation algorithms across various recommendation scenarios [12], [15]. Nevertheless, many existing hybrid recommendation systems primarily focus on user-item similarity and historical interactions while overlooking provider credibility factors such as booking success rates, interaction frequency, and overall provider reliability.

Based on these limitations, this research proposes a hybrid recommendation framework integrating TF-IDF, Multilingual BERT, and Credibility-Based Collaborative Filtering to improve recommendation accuracy and personalization.

III. METHODOLOGY

A. Data Collection and Preparation

Three datasets were used for developing and evaluating the recommendation framework.

The Provider Dataset contains service provider information including provider names, service categories, locations, descriptions, ratings, and experience levels. The User Request Dataset contains customer service requests submitted through the platform. The User Interaction Dataset stores user-provider interaction records including ratings, booking outcomes, and interaction history.

B. TF-IDF Based Content Matching

The TF-IDF module was implemented to capture keyword-level relevance between user requests and provider descriptions.

Initially, provider descriptions were transformed into TF-IDF vectors using a TF-IDF Vectorizer. The vectorizer converts textual information into numerical representations based on term frequency and inverse document frequency statistics [3].

The TF-IDF score is calculated as follows:

$$TFIDF(t,d) = TF(t,d) \times IDF(t) \tag{1}$$

where TF represents term frequency and IDF represents inverse document frequency.

After vectorization, cosine similarity is used to calculate the similarity between the user request and each provider description [3].

$$Similarity = Cosine(Query, Provider) \tag{2}$$

Providers with higher similarity scores are considered more relevant to the user request.

C. BERT-Based Semantic Matching

To overcome the limitations of keyword matching, a semantic similarity module based on Multilingual BERT was integrated into the framework.

The Sentence Transformer model "paraphrase-multilingual-MiniLM-L12-v2" was utilized to generate semantic embeddings for user requests and provider descriptions [1], [2], [13].

The process consists of:

- Encoding user requests into vector embeddings.

- Encoding provider descriptions into vector embeddings.
- Computing cosine similarity between embeddings.
- Generating semantic relevance scores.

Unlike TF-IDF, the BERT module can identify semantic relationships even when different words are used to describe similar services. This capability significantly improves recommendation quality for natural language user requests [1], [2], [6].

D. Custom Credibility-Based Collaborative Filtering

A custom collaborative filtering algorithm was developed as the primary contribution of this research. Unlike traditional collaborative filtering approaches that rely solely on user-item interactions, the proposed method incorporates provider credibility information into the recommendation process [5]. Three datasets consisting of provider information, user requests, and historical user interactions were used to train and evaluate the proposed recommendation framework.

The credibility score is calculated as follows:

$$\begin{aligned} \textit{Credibility} = {} & 0.50(\textit{Rating}) \\ & + 0.30(\textit{SuccessRate}) \\ & + 0.20(\textit{InteractionFrequency}) \end{aligned} \tag{3}$$

The interaction frequency component is transformed using the hyperbolic tangent function to prevent providers with extremely large interaction counts from dominating the ranking process. After calculating credibility scores, Min-Max normalization is applied to convert all scores into a common range between 0 and 1.

User interaction histories are then analyzed to identify providers previously used by each customer. Providers found within a user's interaction history receive a loyalty-based personalization boost.

$$PersonalizedCF = Credibility \times 1.2 \tag{4}$$

This mechanism increases the ranking of providers with whom users have previously interacted while still considering overall provider credibility.

Finally, all scores are clipped to the range [0,1] and missing values are replaced using a neutral score of 0.5 to ensure numerical stability during recommendation generation.

E. Hybrid Score Fusion

The outputs of the TF-IDF, BERT, and Collaborative Filtering modules are combined through a weighted fusion strategy. Before fusion, all component scores are normalized into the range [0,1] to ensure fair comparison and prevent any individual component from dominating the final recommendation process [4], [5], [12].

The final hybrid score is calculated as:

$$HybridScore = 0.30(TFIDF) + 0.35(BERT) + 0.35(CF) \tag{5}$$

The weighting strategy was selected to provide balanced contributions from semantic similarity and collaborative filtering while maintaining content relevance through TF-IDF matching [12], [15]. Providers are ranked according to their hybrid scores, and the Top-20 providers are returned as the final recommendation output.

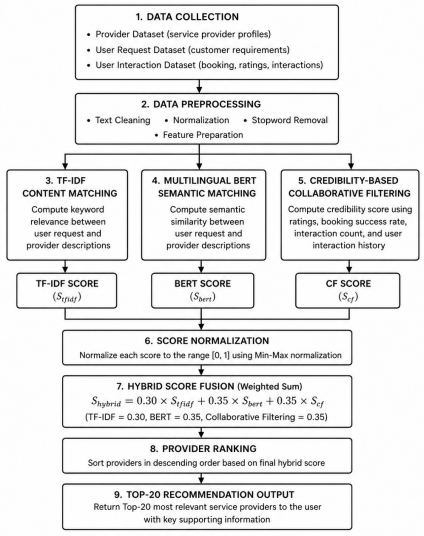


Fig. 1. Hybrid Recommendation Engine Methodology Workflow

IV. SYSTEM DESIGN AND IMPLEMENTATION

A. System Overview

The proposed Hybrid Service Provider Recommendation Framework consists of three recommendation modules integrated through a weighted score fusion mechanism. The system accepts a natural language user request and generates the Top-20 most relevant service providers by combining content-based filtering, semantic similarity analysis, and collaborative filtering. The overall workflow includes data preprocessing, feature extraction, recommendation score generation, score normalization, hybrid fusion, and provider ranking [4], [5], [12].

B. Database Design

MongoDB v6.0 with Vector Search was selected as the primary database because the proposed framework stores both structured provider information and vector embeddings generated by the semantic matching model. The document-oriented architecture of MongoDB allows flexible storage of provider profiles while efficiently supporting similarity search operations while maintaining scalability for large recommendation datasets [4], [5].

C. Software Implementation

The recommendation engine was implemented using Python. Data preprocessing and feature engineering were performed using Pandas and NumPy. TF-IDF vector generation and cosine similarity calculations were implemented using

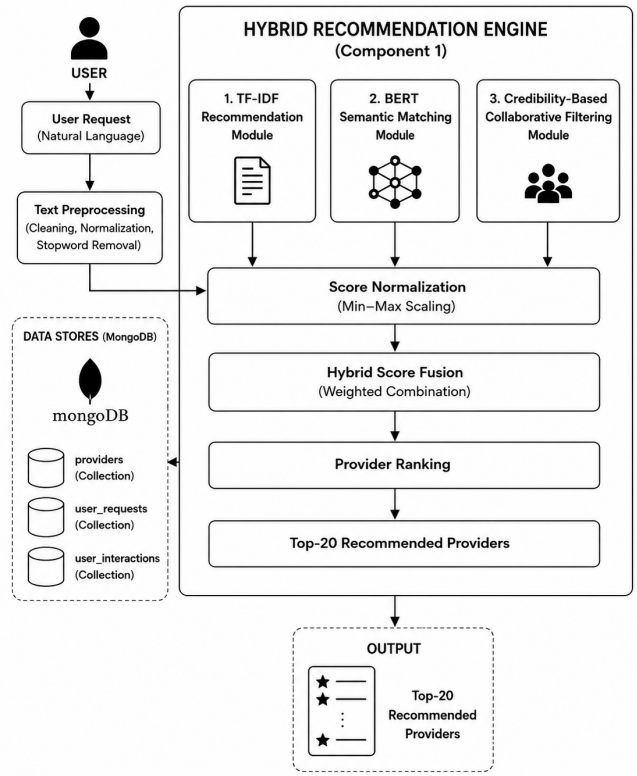


Fig. 2. System Overview Diagram

Scikit-learn [3]. Semantic similarity was implemented using the Sentence Transformers framework with the *paraphrase-multilingual-MiniLM-L12-v2* model [1], [2], [13]. All experiments were executed in Google Colaboratory.

TABLE I
TECHNOLOGIES USED FOR THE RECOMMENDATION COMPONENT

Component	**Technology**
Programming Language	Python 3
Development Platform	Google Colab
Database	MongoDB v6.0 + Vector Search
NLP Library	Sentence Transformers
ML Library	Scikit-learn
Data Processing	Pandas, NumPy

D. Provider Data Management

Provider profiles were stored as MongoDB documents containing service category, provider description, district, years of experience, average rating, and credibility-related attributes. During implementation, textual information was combined into a single feature representation to improve keyword matching and semantic similarity computation [3], [10].

Interaction statistics, including average ratings, booking success rates, and interaction frequencies, were dynamically calculated from the user interaction collection before recommendation generation. These statistics serve as the foundation for computing provider credibility scores and personalized recommendation scores.

E. Recommendation Engine Integration

Unlike conventional recommender systems that execute a single recommendation technique, the proposed implementation executes three recommendation modules independently [4], [5], [15].

- TF-IDF module generates keyword relevance scores.
- Multilingual BERT module computes semantic similarity scores.
- Credibility-Based Collaborative Filtering module produces personalized credibility scores.

Each module produces scores independently, allowing individual evaluation and future replacement without affecting the remaining components. The modular architecture also improves system scalability and simplifies future model enhancements.

F. Score Fusion Implementation

To ensure fair comparison among recommendation modules, all scores are normalized to the range of 0 to 1 using Min-Max normalization before score fusion. The normalized scores are combined using weights of 0.30 for TF-IDF, 0.35 for Multilingual BERT, and 0.35 for Collaborative Filtering. This configuration balances keyword relevance, semantic understanding, and personalized recommendation, while similar weighted hybrid strategies have demonstrated improved recommendation performance [4], [12], [15].

G. Recommendation Output

After hybrid score computation, providers are sorted in descending order according to their final scores. The highest-ranked twenty providers are returned to the user together with supporting information including provider rating, experience, district, and individual component scores.

The modular implementation also allows intermediate scores generated by each recommendation module to be analyzed separately during performance evaluation using standard Information Retrieval metrics such as Precision@K, Recall@K, and NDCG@K [11].

V. RESULTS AND DISCUSSION

The proposed Hybrid Recommendation Framework was evaluated using a large-scale dataset consisting of 10,000 service providers, 20,000 user requests, and 100,000 user-provider interactions. The evaluation was conducted to compare the performance of the proposed hybrid model against its individual recommendation components, namely TF-IDF, Multilingual BERT, and Collaborative Filtering (CF). Standard Information Retrieval (IR) evaluation metrics, including Precision@K and Normalized Discounted Cumulative Gain (NDCG@K), were used to measure recommendation quality [11].

A. Recommendation Performance Evaluation

The recommendation accuracy of the proposed framework was evaluated using Precision@5, Precision@10, Precision@20, NDCG@5, NDCG@10, and NDCG@20. Table III summarizes the experimental results obtained from each recommendation approach.

TABLE II
RECOMMENDATION PERFORMANCE COMPARISON

Method	P@5	P@10	P@20	NDCG@5	NDCG@10	NDCG@20
TF-IDF Only	0.9458	0.9378	0.9231	0.9471	0.9412	0.9297
BERT Only	0.9467	0.9442	0.9387	0.9469	0.9451	0.9410
CF Only	0.0633	0.0651	0.0672	0.0613	0.0633	0.0654
Hybrid	**0.9490**	**0.9483**	**0.9442**	**0.9496**	**0.9490**	**0.9458**

As shown in Table I, the proposed hybrid recommendation framework consistently achieved the highest Precision@K and NDCG@K values across all evaluation settings. Compared with individual recommendation methods, the hybrid approach effectively combines keyword relevance, semantic understanding, and personalized recommendation to generate more accurate provider rankings. Similar improvements have also been reported in previous hybrid recommendation studies [4], [12].

Although the Collaborative Filtering module alone produced relatively lower retrieval performance due to its dependence on user interaction history, its integration within the hybrid framework significantly enhanced personalization without negatively affecting overall recommendation accuracy.

B. Score Distribution Analysis

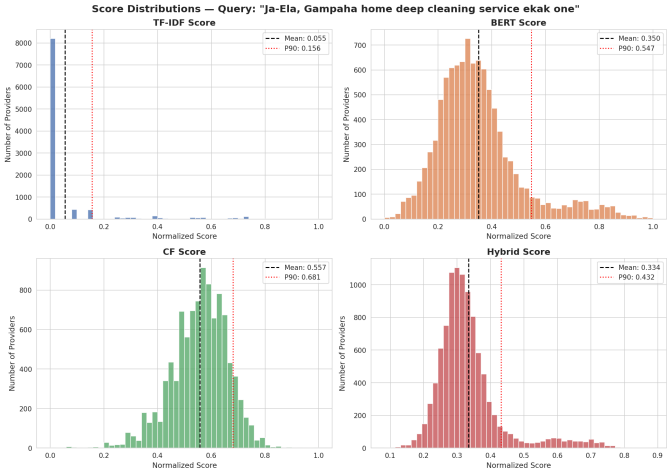


Fig. 3. Score Distributions

Figure 2 illustrates the score distributions produced by the TF-IDF, BERT, Collaborative Filtering, and Hybrid recommendation modules.

The TF-IDF component generated sparse score distributions because it relies on exact keyword matching. In contrast, the BERT module produced smoother score distributions by capturing semantic relationships between user requests and provider descriptions. The Collaborative Filtering module generated credibility-based scores centered around medium-to-high confidence providers based on historical interactions. After weighted fusion, the hybrid scores exhibited a well-balanced distribution, demonstrating that the proposed framework effectively integrates complementary recommendation signals.

C. Component Independence Analysis

Figure 3 presents the Pearson correlation heatmap among the TF-IDF, BERT, Collaborative Filtering, and Hybrid scores.

The correlation analysis shows that the TF-IDF and BERT components exhibit moderate positive correlation, as both evaluate textual relevance using different representation methods. Conversely, the Collaborative Filtering component demonstrates minimal correlation with the content-based modules, indicating that it contributes complementary information derived from user interaction history rather than textual similarity.

The Hybrid score exhibits strong correlation with all three components, confirming that the weighted fusion mechanism successfully integrates their contributions while preserving their individual strengths.

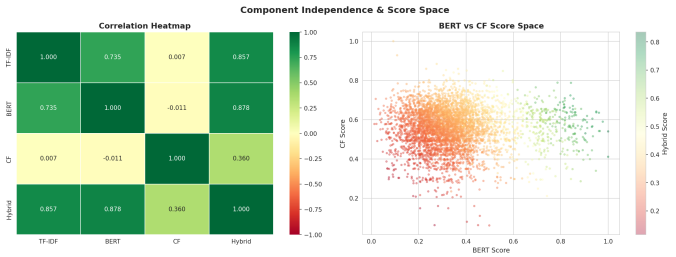


Fig. 4. Pearson Correlation Heatmap

D. Weight Sensitivity Analysis

To evaluate the robustness of the proposed weighted fusion strategy, a sensitivity analysis was conducted by varying the contribution of the BERT component while proportionally adjusting the remaining weights.

The experimental results indicate that recommendation performance remains relatively stable around the selected configuration of 30% TF-IDF, 35% BERT, and 35% Collaborative Filtering. Although minor improvements were observed for slightly different weight combinations, the selected configuration achieved consistently high Precision@20 and NDCG@20 values while maintaining balanced contributions from all recommendation components. This observation is consistent with previous studies on hybrid recommendation systems, where weighted score fusion has been shown to improve recommendation robustness [12], [15].

E. Discussion

The experimental results demonstrate that integrating TF-IDF, Multilingual BERT, and the proposed Credibility-Based

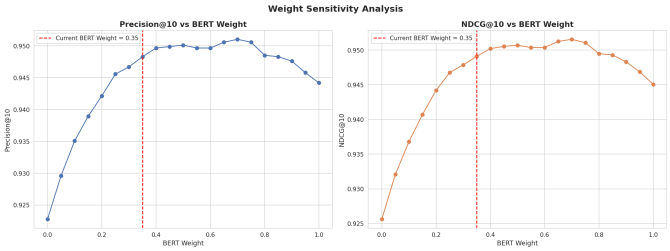


Fig. 5. Weight Sensitivity Analysis

Collaborative Filtering algorithm significantly improves recommendation quality compared with individual recommendation techniques. The weighted hybrid framework consistently achieved higher Precision@K and NDCG@K values while providing accurate semantic matching and personalized provider ranking. These results confirm the effectiveness of combining lexical, semantic, and interaction-based recommendation strategies.

VI. CONCLUSION

This paper presented a Hybrid Recommendation Framework for intelligent service provider matching by integrating TF-IDF, Multilingual BERT, and a custom Credibility-Based Collaborative Filtering algorithm. The proposed framework effectively addresses the limitations of traditional service marketplaces by combining keyword relevance, semantic understanding, and personalized recommendation based on provider credibility and user interaction history [4], [5]. Experimental evaluation using large-scale provider, user request, and interaction datasets demonstrated that the hybrid model consistently outperformed individual recommendation approaches in terms of Precision@K and NDCG@K, which are widely accepted Information Retrieval evaluation metrics for recommendation systems [11]. The weighted fusion strategy successfully balanced the contributions of content-based filtering, semantic similarity, and collaborative filtering, resulting in more accurate and reliable service provider recommendations. Overall, the proposed framework provides an efficient, scalable, and intelligent solution for online service provider recommendation systems while aligning with recent advances in hybrid recommender system research [12], [15].

Future work will focus on incorporating temporal context, real-time provider availability, geographical proximity, and dynamic user preferences to further improve recommendation quality. Additionally, advanced deep learning ranking models and reinforcement learning techniques will be investigated to enhance personalization and adapt recommendations based on evolving user behavior.

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